Fiscal Year:	FY 2021	**Task Last Updated:**	FY 10/18/2020
PI Name:	Yu, Nan Ph.D.		
Project Title:	JPL Participation in ESA ACES (Atomic Clock Ensemble in Space) Worldwide Clock Comparison Campaign		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Fundamental Physics Rapid Turn Around Flight Opportunity
Start Date:	10/01/2010	**End Date:**	05/31/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Changed JPL Task Plan number to 71-15884 per U. Israelsson/JPL (Ed., 5/27/2020) NOTE: End date changed to 5/31/2023 per U. Israelsson/JPL (Ed., 1/28/2020) NOTE: End date changed to 11/29/2019 per PI (Ed., 2/13/18) NOTE: End date changed to 3/31/2018 per PI (Ed., 2/19/15)		
Key Personnel Changes/Previous PI:	For the period of FY2019 and part of FY2020: Key personnel during the reporting period includes Luca Mazzarella, Makan Mohageg, Eric Burt, Robert Tjoelker, Vladmir Schkolnik, Jason Williams, and James Kohel		
COI Name (Institution):	Tjoelker, Robert Ph.D. (Jet Propulsion Laboratory) Burt, Eric Ph.D. (Jet Propulsion Laboratory)		
Grant/Contract No.:	JPL Task Plan number 71-15884 ; JPL Task Plan number 71-15743		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Tremendous progress has been made recently in the U.S. and other countries in advancing atomic clock technology and its applications for the benefit of science, national interests, and society in general. Space provides a unique environment in which many benefits can be realized. The European Space Agency (ESA) is readying the ACES (Atomic Clock Ensemble in Space) payload for the International Space Station (ISS) for an anticipated 2013 launch [Ed. note Oct 2020: anticipated launch 2021]. Some of the key objectives of ACES will be achieved by global time transfer and clock comparisons with the most advanced clocks at multiple worldwide sites. There is an opportunity for investigators from the leading U.S. clock and metrology labs to participate in the clock comparison campaign, enhancing the overall science objective of ACES and at the same time demonstrating and providing science benefit from advanced U.S. clock technology. The Jet Propulsion Laboratory (JPL) is one of the major U.S. time and frequency metrology institutions serving NASA missions and Deep Space Network needs. It was the NASA center for the microgravity LCAP program (Laser Cooling and Atomic Physics) in charge of developing the cold atom clock payload for PARCS (Primary Atomic Reference Clock in Space). JPL has also developed a unique atomic frequency standard that has demonstrated an exceptional long-term stability. We propose to collaborate with ESA on the ACES mission by participating in the ACES global clock campaign with JPL's ultra-stable mercury ion clock. During a proposed six-year performance period, we will: 1) contribute significantly to the ACES mission science objectives by providing unique clock references from a Western U.S. site; 2) provide and demonstrate a high-impact application for JPL's high performance time keeping clock technology; and 3) re-establish NASA's involvement in ISS Clock Science and initiate a healthy collaboration between NASA and ESA in the area of clock and quantum sensors. The task is to collaborate with ESA on the ACES mission by participating in the ACES global clock comparison campaign and provide frequency and time references derived from JPL's ultra-stable mercury ion clock. These references will be connected to a microwave link ground terminal provided by ACES and installed at JPL. Clock comparisons will be carried out over the microwave link with ESA clocks on the ISS. As JPL is the only ground comparison site located in the western USA and the mercury ion clock is the only one of its kind in the global clock comparison campaign, JPL's participation will contribute significantly to the overall ACES mission science objectives, while simultaneously providing a high-impact application for JPL's high performance time keeping clock technology. The task will also help re-establish NASA's involvement in ISS clock science and enable future collaboration between NASA and ESA in the area of clocks and quantum sensors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Advanced atomic clocks are critical to the interests of NASA and the nation. Telecommunication and navigation (e.g., GPS) are well-known applications of space clocks. It is also part of quantum technology development programs. Within NASA, high performance atomic clocks and frequency and timing capabilities enable operation of the NASA Deep Space Network and provide fundamental tools for precise measurements exploring the laws of Nature. Progress in advanced clock technology development, both on ground and for space, is rapidly being made in many countries. Participation by the U.S. in the ACES mission contributes to the science objectives of the ISS consistent with the 2011 National Research Council (NRC) report recommendations in Fundamental physics and demonstrates the capability and potential of NASA mercury ion clock technology. It will also help advances the clock development and operational capabilities at JPL/NASA.
Task Progress:	The JPL's efforts have been in three main areas while anticipating the ACES payload launch: maintaining and upgrading the JPL ground clocks, preparing for the installation of the microwave link ground terminal at JPL site, and conducting investigations of ACES clock science. In the past year, the focus has been on ground clocks and science investigations. The ground terminal site preparation has been mostly completed and is ready to receive the ESA ground terminal to be installed at JPL. The ground clock upgrading efforts have since been shifted to the highly leveraged development of higher performance optical clocks, both a Sr lattice clock and a single trapped Yb+ clock, which will officer increased values to the ACES global clock comparison experiments. While the Sr lattice optical clock experiment is now at its final stage of demonstrating clock transitions, the ion clock is being built up. Meanwhile we have continued developing a science data analysis tool kit using the simulated data from the ACES project data analysis center. Programmatically, the ACES payload launch date is now scheduled around August 2021, and the ground terminal delivery has been delayed until after the ACES launch.
Bibliography Type:	Description: (Last Updated: 10/28/2020)
Papers from Meeting Proceedings	Schkolnik V, Williams JR, Yu N. "Direct semiconductor diode laser system for an optical lattice clock based on neutral strontium for future tests of fundamental physics in space." SPIE 11296, Optical, Opto-Atomic, and Entanglement-Enhanced Precision Metrology II, San Francisco, California, February 1-6, 2020. Proceedings Volume 11296, Optical, Opto-Atomic, and Entanglement-Enhanced Precision Metrology II; 112962H. 2020. https://doi.org/10.1117/12.2549377 , Feb-2020